

CMPG 325 – Individual Semester Project

Milestone 1

Client Design Review

Network Design Project

Ditso Dairy Farm (Bloemhof)

Bloemhof, North West Province, South Africa

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Milestone 1 Submission Package

1. Client Requirements
2. Design Constraint
3. Addressing Block
4. Change Request CR2
5. Devices
6. Physical Topology
7. Logical Topology
8. Mapping of Design Decisions to Functional Requirements

The ninth component, the Initial GitHub Repository, accompanies this document as a separate submission link.

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1 Client Requirements — and How Each Is Satisfied

Ditso Dairy Farm is a working dairy operation based in Bloemhof, North West Province, and is best understood as a single-site agricultural client with an office/administration function, day-to-day farm operations (milking parlour and herd management), and a modest need for guest or vendor network access. The network design below is built directly and only around the requirements set out in the approved client brief (CMPG325-2026-039).

Requirement	Detail from the Brief	How the Design Satisfies It
Assigned organisation & industry	Ditso Dairy Farm (Bloemhof), Agriculture	VLAN and device naming reflects real farm functions – office/admin, farm operations (milking parlour), management/infrastructure, and guest/vendor access – rather than a generic office layout.
Assigned addressing block	10.21.0.0/16	All subnets are carved exclusively from this block (Section 3); nothing is borrowed from outside it.
Assigned technical challenge	VLANs (switch-based segmentation design), Intermediate difficulty	Six VLANs are created, each its own broadcast domain, switched at access/core level and routed centrally by R1 using router-on-a-stick sub-interfaces (Section 7).
Appropriate connectivity & network services	Provide appropriate connectivity and network services for the assigned scenario	Full inter-VLAN routing is provided via R1, with DHCP service scoped per VLAN so end devices obtain addressing automatically.
Design constraint	No network downtime tolerated during month-end processing	Addressed through VLAN isolation, access-port hardening and a documented change-freeze policy (Section 2).
Change request	CR2: client takes over an additional floor/area and needs coverage there	A single new access switch and VLAN extend coverage to the new area without altering the existing design (Section 4).
Working, testable Packet Tracer implementation	Produce a working, testable Packet Tracer implementation	The topology uses only standard Packet-Tracer-supported devices and IOS commands, so it can be rebuilt, opened and re-tested reliably.
Design & configure appropriate topology, routers, switches and end devices	Use Cisco Packet Tracer; design an appropriate topology; configure the necessary routers, switches, end devices and other required nodes	See device inventory (Section 5), physical topology (Section 6) and logical topology (Section 7).
Demonstrate connectivity & document evidence	Demonstrate successful connectivity and testing; document significant design decisions and evidence in GitHub	This document itself is written for inclusion in the project's GitHub portfolio, with each design decision justified in place rather than left implicit.

2 Design Constraint — “No Network Downtime Tolerated During Month-End Processing”

Month-end processing (billing and farm records) takes place on the **Admin VLAN**. Because the assigned technical challenge for this project is VLANs at Intermediate difficulty – not a High-Availability or redundancy challenge – the constraint is addressed primarily through segmentation and operational discipline rather than through additional hardware, which keeps the solution matched to the brief’s stated scope.

- **VLAN segmentation as the primary defence.** Admin traffic is isolated in its own broadcast domain, so congestion, a misconfiguration, or a misbehaving device on the Farm-Ops, Guest or New-Area VLANs cannot disrupt Admin traffic during month-end.
- **PortFast and BPDU Guard on all access ports.** Prevents an accidental topology change (e.g. a device plugged into the wrong port, or a switching loop) from causing an outage on any VLAN, Admin included.
- **Change-freeze policy.** VLAN, trunk and routing changes are scheduled outside the month-end processing window. This is a documented operational decision rather than a configuration line, and it is recorded here and in the GitHub README as a deliberate design choice.
- **Configuration backup discipline.** `copy running-config startup-config` is only run once a change has been verified as working, so that a bad change can be rolled back quickly rather than persisting through a reload.

Documented enhancement (not implemented at this milestone). A second uplink from each access switch to the core switch, with Spanning Tree Protocol handling automatic failover, would add further protection against a single cable or port failure. This is recorded here as a considered extension rather than built into the base design, since introducing dual-core redundancy would extend the project beyond the Intermediate-level VLAN challenge that was actually assigned, and the brief cautions against introducing additional scope beyond what a requirement calls for.

3 IP Addressing Plan

Assigned Network Block: 10.21.0.0/16

Subnetting Methodology: Fixed /24 subnets are utilized to provide straightforward segmentation, simplify documentation, and allow substantial headroom for future agricultural or sensor network expansion.

3.1 VLAN and Subnet Allocation

VLAN	Segment Name	Network Subnet	Default Gateway	Purpose
10	ADMIN	10.21.10.0/24	10.21.10.1	Office, billing, and month-end records
20	FARM-OPS	10.21.20.0/24	10.21.20.1	Milking parlour systems and herd terminals
30	MGMT-SERVERS	10.21.30.0/24	10.21.30.1	Farm management server and IT admin PC
40	GUEST	10.21.40.0/24	10.21.40.1	Visitor and vendor network access

VLAN	Segment Name	Network Subnet	Default Gateway	Purpose
50	NEW-AREA-CR2	10.21.50.0/24	10.21.50.1	Coverage for additional floor/area (CR2)
99	NATIVE-MGMT	10.21.99.0/24	10.21.99.1	Switch management only; no user traffic

3.2 Detailed Subnet Calculation Reference

VLAN	Network	Mask	Usable Host Range	Broadcast	Gateway
10	10.21.10.0	/24	10.21.10.1 – 10.21.10.254	10.21.10.255	10.21.10.1
20	10.21.20.0	/24	10.21.20.1 – 10.21.20.254	10.21.20.255	10.21.20.1
30	10.21.30.0	/24	10.21.30.1 – 10.21.30.254	10.21.30.255	10.21.30.1
40	10.21.40.0	/24	10.21.40.1 – 10.21.40.254	10.21.40.255	10.21.40.1
50	10.21.50.0	/24	10.21.50.1 – 10.21.50.254	10.21.50.255	10.21.50.1
99	10.21.99.0	/24	10.21.99.1 – 10.21.99.254	10.21.99.255	10.21.99.1

4 Change Request CR2 — Additional Floor/Area

Wording of the request: “*The client takes over an additional floor/area of the building and needs coverage there.*” The brief is explicit that the final solution must show how the network accommodates this change *rather than introducing additional scope*, so the response below is deliberately minimal and additive.

How it is satisfied: a single new access switch, SW-NEWAREA, is added as a new leaf off the existing core switch, carrying its own VLAN (50) and subnet (10.21.50.0/24, Section 3). No change is required to the router configuration beyond adding one sub-interface, no change is required to any existing VLAN, and the only change at the core switch is allowing VLAN 50 on the trunk toward SW-NEWAREA. This is exactly the kind of scalability an extended-star topology is intended to demonstrate: the new area is served without redesigning anything already in place.

5 Devices

Device counts were kept deliberately modest and realistic for a single-site dairy farm with a small administrative and technical footprint, consistent with the brief’s instruction to design specifically for the assigned client rather than a larger or more elaborate scenario.

Area	VLAN	Devices	Count
Admin Office	10	Office/admin PCs (billing, records) + shared network printer	3 PCs + 1 printer
Milking Parlour / Farm Operations	20	Herd-management terminal + milking-parlour control PCs	3

Area	VLAN	Devices	Count
Server / Equipment Room	30	Farm management server + IT administration PC (attached directly to the core switch)	2
Guest / Vendor Access	40	Wireless access point for visitor/vendor devices + one wired vendor-facing PC	2
New Floor/Area – CR2	50	Additional office PCs for the newly occupied area	3
Infrastructure	99	R1 (router), SW-CORE, and four access switches (SW-ADMIN, SW-FARMOPS, SW-GUEST, SW-NEWAREA) – no end-user devices	1 router + 5 switches

In total, the design provisions for **13 end-user devices** across four access switches, plus the farm management server and IT PC attached directly to the core switch in the server/equipment room, and six infrastructure devices (one router, one core switch, four access switches).

6 Physical Topology

Topology type: Extended Star (two-tier hierarchical – distribution and access). R1 and SW-CORE form the distribution point in the server/equipment room; SW-ADMIN, SW-FARMOPS, SW-GUEST and SW-NEWAREA are access-layer leaves serving their respective areas. This is the appropriate choice here because it scales cleanly (adding SW-NEWAREA for CR2 required zero redesign of the rest of the network), isolates physical faults so that a cable or switch failure in one area cannot affect another, and is a standard, PT-native topology at Intermediate difficulty.

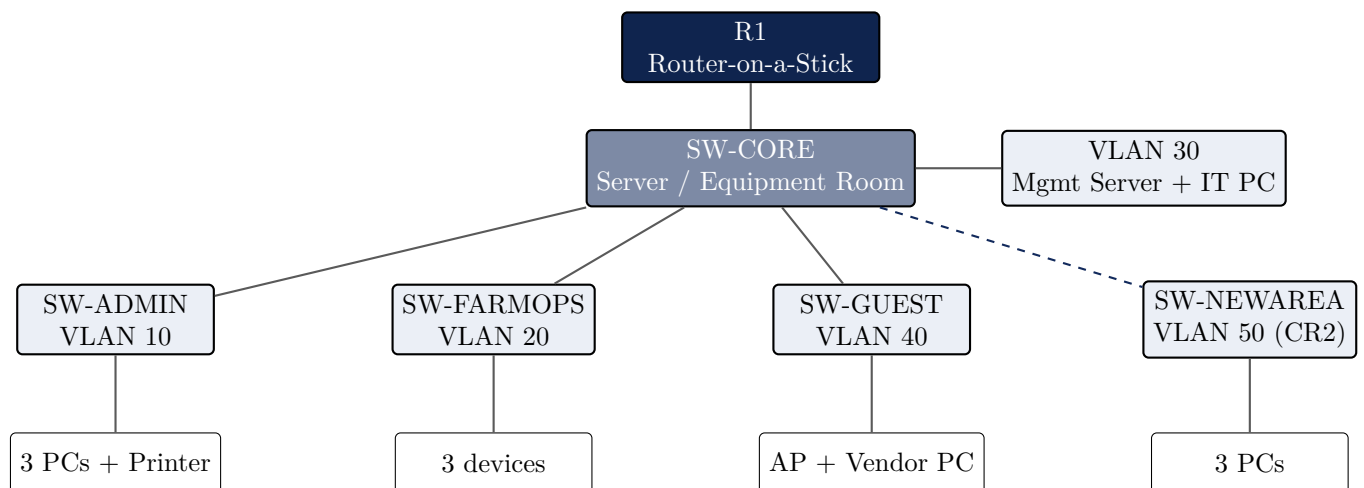


Figure 1: Physical topology of the Ditso Dairy Farm network. The dashed link marks the switch and VLAN added for Change Request CR2.

Cabling summary: the Router-to-Core link is a single trunk carrying all VLANs. Core-to-Access links are trunk links carrying tagged VLAN traffic for the relevant area only. Access-to-end-device links are standard access-mode ports, each assigned to a single VLAN. The management/server

devices in VLAN 30 connect directly to access ports on the core switch, since they are physically housed in the same restricted server/equipment room as the core infrastructure.

7 Logical Topology

The logical topology shows how traffic flows between VLANs independently of physical location. R1 uses sub-interfaces (router-on-a-stick) to route between all six VLANs over the single physical trunk link to SW-CORE. Because each VLAN is a separate broadcast domain, the only path between them is through R1's Layer-3 routing – this is the actual demonstration of the assigned VLAN (switch-based segmentation) challenge.

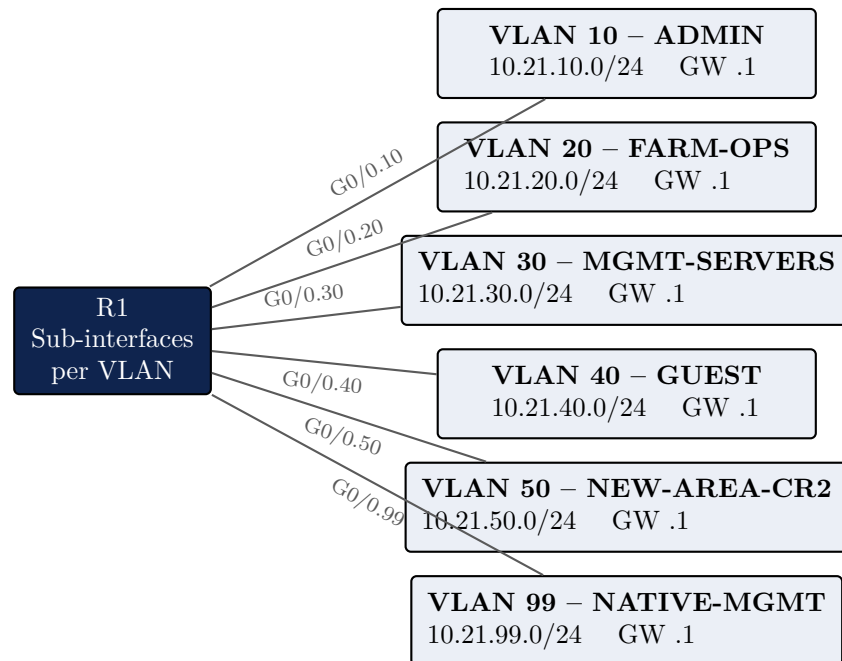


Figure 2: Logical topology showing VLAN-to-subnet mapping via router sub-interfaces.

8 Mapping of Design Decisions to Functional Requirements

Functional Requirement	Addressed By
Segmentation (VLANs, switch-based)	Six VLANs, each its own broadcast domain, aligned to a distinct farm function rather than to physical rooms alone.
Efficient, documented addressing	Six /24 subnets carved from the assigned 10.21.0.0/16 block, fully tabulated in Section 3, with over 250 subnets reserved for future growth.
No downtime during month-end processing	Admin VLAN isolation, PortFast/BPDU Guard on access ports, a documented change-freeze policy, and disciplined configuration backups (Section 2).
Change Request CR2	One new access switch and one new VLAN/subnet extend coverage to the new area with no change to existing VLANs or devices (Section 4).
Working, testable Packet Tracer implementation	Standard Packet-Tracer-supported hardware and IOS features only, so the .pkt file opens and reproduces the working solution reliably.

Functional Requirement	Addressed By
Documentation of design decisions	Every decision in this document is paired with its justification, ready for inclusion in the project's GitHub portfolio.